CHAPTER 10:
SEQUENCES AND SERIES

In this unit, we will introduce the concept of both finite and infinite sequences and series.

SEQUENCES

A *sequence* is a list of terms indexed by positive integers.
$\{a_n\}_{n=1}^{\infty}$
a: $N \rightarrow \mathbb{R}$, where N = {1,2,3,...}

A sequence can often be represented as a discrete function.

In layman's terms, a sequence is a list of numbers that often have a particular order. They can have an finite or infinite number of terms, in which case it goes on forever (represented by an ellipsis ...)

Like sets, sequences are written by listing each element, separated by a comma and enclosed by curly brackets:
{ 2, 5, 8, 11}

If the sequence is infinite, simply denote it by adding an ellipsis (...) following the last term:
{ 2, 5, 8, 11 ... }

In most cases, you will see sequences represented in a form that looks like this:
a_n = {1, 2, 3, ...n, (n+1)... }

a_n refers to the *term* in the sequence
n, or the subscripted variable, is the *term number* in the sequence.

For example, let $a_n = \{1, 2, 4, 8, 16\}$

a_3 is the 3rd term in this sequence, which is 4.

Let $b_n = \{0, 1, 0, 1, 0\}$

b_1 is the 1st term in this sequence, which is 0. b_5 is the 5th term in this sequence, which also happens to be 0.

Unlike sets, elements in sequences can repeat indefinitely!

Most examples of sequences that we see will have some sort of pattern to the enumeration of the numbers. For instance, take a look at the following:

$a_n = \{2,\ 4,\ 6,\ 8,\ 10,\ 12,\ \ldots\}$

$a_n = \{3,\ 1,\ \frac{1}{3},\ \frac{1}{9},\ \frac{1}{27},\ \frac{1}{81},\ \ldots\}$

$a_n = \{1,\ 3,\ 8,\ 15,\ 24,\ 35,\ 48\}$

For arithmetic and geometric sequences, we can determine the pattern and write an explicit formula to determine any a_n in the sequence given certain information (for instance, the value of the starting term).

SERIES

A *series* is a summation of a list of consecutive terms in a sequence.

For example, let $a_n = \{a_1, a_2, a_3, \ldots a_N\}$

$$\sum_{n=1}^{N} a_n = a_1 + a_2 + a_3 + \ldots a_N$$

$\sum_{n=i}^{N} a_n$ is referred to as *sigma notation,* and is used commonly when denoting a series.

n is what we call a *dummy variable*: it's more of a placeholder if anything, and represents the *term number*. Norte that *n* could easily be replaced by another variable without the slightest change to the expression. For future reference, we will be using *i* and *n* interchangeably as our dummy variable.

N represents the upper bound for the *term number* -- if the sequence is finite, *N* will be some nonnegative integer $\mathbb{Z}$. However, if the sequence is infinite, then *N* will have the "value" of $+\infty$.

i is the starting index. In *most* cases, this starting index will be 1.

Here we must also make a clear distinction between *finite* and *infinite* series. A finite series $\sum_{i=1}^{n} a_i$ sums the terms in the sequence a_n to a finite index N; that is, the N th term in the sequence.

For instance, let $a_n = \{a_1, a_2, a_3, \ldots a_{7, \ldots}\}$. The series $\sum_{n=i}^{7} a_n$ would be the sum of the first 7 terms in the sequence; that is, $\sum_{n=1}^{7} a_n = a_1 + a_2 + a_3 + \ldots a_7$.

An *infinite series*, in comparison, sums the terms in an infinite sequence; in other words, there are an infinite number of terms added.

Let $a_n = \{a_1, a_2, a_3, \ldots a_{N, \ldots}\}$. The series $\sum_{n=i}^{\infty} a_n$ is essentially an infinite sum of all the terms in a_n;

that is, $\sum_{n=i}^{\infty} a_n = a_1 + a_2 + a_3 + \ldots a_N + \ldots$

It is clear that the sum of a finite number of terms would yield a finite answer. However, although perhaps counterintuitive, the sum of an infinite series can likewise converge to a finite answer.

The latter value is, in essence, the limit of the Nth partial sum of the series.

If the limit exists, then $\sum_{n=i}^{\infty} a_n = \lim_{N\to\infty} \sum_{n=i}^{N} a_n$.

<u>To Summarize:</u>

For an infinite series $\sum_{n=1}^{\infty} a_n$, the nth partial sum is given by

$S_n = a_1 + a_2 + a_3 + \ldots + a_n$.

If $\lim_{n\to\infty} \{S_n\}$ converges to some L, then the infinite series converges.

If $\lim_{n\to\infty} \{S_n\}$ diverges, then the infinite series diverges.

But how do we know if, given an infinite series, that series converges or diverges?

That is what we will examine in the rest of this chapter.

<u>WARNING!</u>

We will now proceed to cover the convergence tests for infinite series. Although we will not discuss the proofs of each theorem, it is integral (pun not intended) that you understand each test and can apply them to series - without being told which test to use. The first two series are "special cases", and are fairly easy to recognize. Unfortunately, the same cannot be said true for many of the other tests - we highly recommend you do a lot of practice problems to better prepare you for the AP ® Calculus exam.

So without further ado:

<u>"Special Cases"</u>

- Geometric Series
- Telescoping Series

<u>Convergence Tests for Nonnegative Series</u>

- nth Term Test (Divergence)
- Integral Test
- P-Series Test
- Root Test
- Ratio Test
- Direct Comparison Test
- Limit Comparison Test

<u>Alternating Series</u>

- Alternating Series Test

Geometric Series

A geometric series can be written in the form $\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + ar^3 + \ldots ar^{n-1} + \ldots$

This should be a review for Precalculus: A geometric sequence is one such that there exists a common ratio between 2 consecutive terms.

For instance, $a_n = \{1, 2, 4, 8, 16\}$ is a geometric sequence. We can writ it's explicit form as $a_n = 2\,(2)^{n-1}$
In general, the explicit form of a geometric sequence can be written as $a_n = a_1(r)^{n-1}$, where a_1 is the first term in the sequence and r is the common ratio.

A geometric *series* $\sum_{n=1}^{\infty} ar^{n-1}$ converges if and only if $|r| < 1$. If S_n converges, then $S_n = \frac{a_1}{1-r}$.

Why $\frac{a_1}{1-r}$? Recall from Precalculus that the partial sum of a geometric sequence $S_n = \frac{a_1(1-r^n)}{1-r}$. If $|r| < 1$, then $\lim_{n\to\infty} \frac{a_1(1-r^n)}{1-r} = \frac{a_1(1)}{1-r}$ since raising a number on the interval $[-a, a]$ by an infinitely large quantity result in an infinitely small number.

Let's take a look at a few examples:

1. $\sum_{n=1}^{\infty} (\frac{1}{2})^{n-1}$ converges since $|\frac{1}{2}| < 1$. Since $a_1 = 1$, $\sum_{n=1}^{\infty} (\frac{1}{2})^{n-1} = \frac{1}{1-\frac{1}{2}} = 2$.

2. $\sum_{n=1}^{\infty} (-\frac{4}{3})^{n-1}$ diverges since $|-\frac{4}{3}| < 1$.

3. $\sum_{n=1}^{\infty} \frac{1}{3^n}$ can be rewritten as $\sum_{n=1}^{\infty} (\frac{1}{3})^n$, which better resembles the form that we are used to.

Since $|\frac{1}{3}| < 1$, we know that $\sum_{n=1}^{\infty} \frac{1}{3^n}$ converges to $\frac{\frac{1}{3}}{1-\frac{1}{3}} = \frac{1}{2}$.

4. $\sum_{n=1}^{\infty} 9^{-n+2}4^{n+1}$ might not at first look like a geometric series, but it can be rewritten as follows:

$\sum_{n=1}^{\infty} 9^{-n+2}4^{n+1} = \sum_{n=1}^{\infty} 9^{-(n-2)}4^{n+1} = \sum_{n=1}^{\infty} \frac{4^{n+1}}{9^{n-2}} = \sum_{n=1}^{\infty} \frac{4^n(4)}{9^n(9^{-2})} = \sum_{n=1}^{\infty} 324(\frac{4}{9})^n$. This is a geometric series where $r = \frac{4}{9}$ and $a_1 = 144$.

$\sum_{n=1}^{\infty} 324\,(\frac{4}{9})^n$ converges since $|\frac{4}{9}| < 1$. Since $a_1 = 144$, $\sum_{n=1}^{\infty} 324(\frac{4}{9})^n = \frac{144}{1-\frac{4}{9}} = \frac{1296}{5}$.

Telescoping Series

A telescoping series is a series whose partial sums cancel to a finite number of terms. This is formally known as the *method of differences*.

Given a telescoping series, we can cancel out the middle terms until we are left with only the initial and final term. From there, we simply take the limit of the partial sum in order to determine whether or not the series converges or diverges.

For example, take a look at the following series:

1. $\sum_{n=1}^{\infty} \frac{1}{n} - \frac{1}{n+1}$

We can rewrite this series as the limit of a partial sum as n approaches infinity:

$$\sum_{n=1}^{\infty} \frac{1}{n} - \frac{1}{n+1} = \lim_{N\to\infty} \sum_{n=1}^{N} \frac{1}{n} - \frac{1}{n+1}$$

By writing out a few of the terms, we can see that the middle terms will cancel out each other, leaving only the first term and last term intact:

$$\lim_{N\to\infty} \sum_{n=1}^{N} \frac{1}{n} - \frac{1}{n+1} = \lim_{N\to\infty} \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{5}\right) + \ldots + \left(\frac{1}{N} - \frac{1}{N+1}\right)$$

Now we simply take the limit of the partial sum to determine what the series converges to:

$$\lim_{N\to\infty} 1 + \left(\frac{1}{N} - \frac{1}{N+1}\right) = 1$$

Sometimes, however, it may not be clear that the series in question is a telescoping series. In the example below, we must first decompose the fraction into 2 separate fractions, from at that stage it is easy to tell that the series is telescoping:

$$\sum_{n=0}^{\infty} \frac{1}{n^2+3n+2} = \lim_{N\to\infty} \sum_{n=1}^{N} \frac{1}{n+1} - \frac{1}{n+2} = \lim_{N\to\infty} \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \ldots + \left(\frac{1}{N+1} - \frac{1}{N+2}\right) = \lim_{N\to\infty} \frac{1}{2} + \left(\frac{1}{N+1} - \frac{1}{N+2}\right) = \frac{1}{2}$$

For these sequences - geometric and telescoping, if they converge, we can easily find the value to which it converges.

However, this is not true for some of the other tests that we are about to cover. For most of those tests, we cannot determine the value to which the sum converges, but rather whether or not it converges in the first place. But fortunately, that's all you really need to know for the AP Calculus ® exam.

Onwards!

<u>CONVERGENCE TESTS FOR NONNEGATIVE SERIES:</u>

A series $\sum a_n$ is nonnegative if $a_n \geq 0$ for all n.

<u>The Nth Term Test</u>

For a nonnegative series $\sum_{n=1}^{\infty} a_n$, if $\lim_{n\to\infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ diverges.

This is fairly intuitive - if the sequence itself doesn't eventually converge to 0, there is no way that the sum of the terms in the sequence will ever reach a finite value.

Let's take a look at an example:

1. $\sum_{n=1}^{\infty} \frac{2n^2+7n+12}{3n^2}$ diverges since $\lim_{n\to\infty} \frac{2n^2+7n+12}{3n^2} = \frac{2}{3}$.

However, this test cannot prove whether or not a series *converges* - a series could "pass" the nth term test and still diverge nevertheless. In other words, $\lim_{n\to\infty} a_n = 0$ does not necessarily imply $\sum_{n=1}^{\infty} a_n$ converges.

A famous example of such a series is $\sum_{n=1}^{\infty} \frac{1}{n}$, better known as the *harmonic series*.

The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges despite $\lim_{n\to\infty} \frac{1}{n} = 0$.

We must use another test to show that $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

Although it cannot be used to show that a series converges, the nth term test is a quick way to see if a series diverges. If it diverges, then there is no need to continue testing the series; however, if it "passes" the test, it is crucial that one attempt to prove its convergence (or possibly divergence) using another test.

<u>The Integral Test</u>

Let $a_n = f(n)$ for some continuous, positive decreasing function on the interval $[N, \infty)$.

$\sum_{i=N}^{\infty} a_n$ converges if and only if $\int_N^{\infty} f(x)\, dx$. Likewise, if $\int_N^{\infty} f(x)\, dx$ diverges, then $\sum_{n=N}^{\infty}$ *must* also diverge.

Remember the harmonic series? We can use the Integral Test to show that it diverges.

First, however, we need to check to see if the series $\sum_{n=1}^{\infty} \frac{1}{n}$ satisfies all the conditions for the Integral Test to hold.

Is $f(x) = \frac{1}{x}$ continuous on $[1, \infty)$? Check.

Is $f(x) = \frac{1}{x}$ positive for all values $x \in [1, \infty)$? Check.

Finally, is $f(x) = \frac{1}{x}$ decreasing on $[1, \infty)$? Also check.

Great! Now we can use the Integral Test!

First, we see that the integral $\int_1^{\infty} \frac{1}{x}\, dx$ diverges on the aforementioned interval:

$$\int_1^{\infty} \frac{1}{x}\, dx = \lim_{b\to\infty} ln\,(x)\,|_1^b = \lim_{b\to\infty} ln\,(b) = \infty.$$

Since $\int_1^{\infty} \frac{1}{x}\, dx$ diverges on $[1, \infty)$, by the Integral Test, we know that $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges as well.

This test should usually be your last resort because it takes more time than other tests. You should be able to spot series that could be dealt with using this test because usually they look kind of integrable and can't be dealt with any other way. An example is $\sum_{n=1}^{\infty} \frac{ln\,(x)}{x}$. From a quick glance, you can tell that we can do this with u-substitution, and there doesn't seem to be any other way to deal with this.

Let's take a look at a few more examples:

1. $\sum_{n=1}^{\infty} \frac{3n}{n^2+1}$ diverges since $\int_1^{\infty} \frac{3x}{x^2+1}\, dx = \lim_{b\to\infty} \frac{3}{2} ln\,(x^2+1)\,|_1^b = \lim_{b\to\infty} \frac{3}{2} ln\,(b^2+1)\,| = \infty.$

2. $\sum_{n=1}^{\infty} \frac{n}{e^n}$. Using the Integral Test, we can show that this series converges as follows:

$$\int_1^{\infty} \frac{x}{e^x}\, dx = \int_1^{\infty} x \cdot e^{-x}\, dx = -xe^{-x}|_1^{\infty} + \int_1^{\infty} e^{-x}\, dx = \lim_{b\to\infty} -\frac{x}{e^x} - e^{-x}|_1^b = \lim_{b\to\infty} -\frac{b}{e^b} - \frac{1}{e^b} - (-\frac{1}{e} - \frac{1}{e}) = \lim_{b\to\infty} -\frac{b}{e^b} + \frac{2}{e}$$

Notice how we get an indeterminate form, $\frac{\infty}{\infty}$ when we compute the limit $\lim_{b\to\infty} -\frac{x}{e^x}$. So how can we determine the value of the limit? Fortunately, we have L'Hopital's Rule.

Applying L'Hopital's Rule, it follows that:
$\lim_{b\to\infty} -\frac{x}{e^x} = \lim_{b\to\infty} -\frac{1}{e^x} = 0$.

Thus, $\int_1^\infty \frac{x}{e^x}\,dx$ converges to $\frac{2}{e}$, and by the Integral Test, $\sum_{n=1}^{\infty} \frac{n}{e^n}$ *converges.*

It should be noted that $\int_N^\infty f(x)\,dx$ does not necessarily equal $\sum_{i=N}^{\infty} a_n$. The Integral Test only tells one that the series converges, and not necessarily what it converges to.

<u>The P-Series Test</u>

A generalization of the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ is the *p-series*, of the form $\sum_{n=1}^{\infty} \frac{1}{n^p}$, $p \in \mathbb{R}$.

A *p-series* $\sum_{n=1}^{\infty} \frac{1}{n^p}$ diverges if $p \le 1$, and converges if $p > 1$

We have already established that $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges by the Integral Test. Suppose we use the Integral Test to determine whether or not $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.

Since $f(x) = \frac{1}{x^2}$ is continuous, positive and decreasing on $[1, \infty]$, we can proceed as follows:

$$\int_1^\infty \frac{1}{x^2}\,dx = \lim_{b\to\infty}\int_1^b \frac{1}{x^2}\,dx = \lim_{b\to\infty} -\frac{1}{x}\Big|_1^b = \lim_{b\to\infty} -\frac{1}{b} + 1 = 1 \text{ . Thus. } \sum_{n=1}^{\infty} \frac{1}{n^2} \text{ converges.}$$

Now suppose we used the same test for $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{2}}}$:

$$\int_1^\infty \frac{1}{x^{\frac{1}{2}}}\,dx = \lim_{b\to\infty}\int_1^b \frac{1}{x^{\frac{1}{2}}}\,dx = \lim_{b\to\infty} 2x^{\frac{1}{2}}\Big|_1^b = \lim_{b\to\infty} 2\,(b)^{\frac{1}{2}} - 2 = \infty \text{ . Thus, } \sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{2}}} \text{ diverges.}$$

For any $p \le 1$, the integral of the corresponding function diverges. For $p > 1$, the integral of the corresponding function converges. As such, from the Integral Test we can derive the *p-series Test.*

<u>The Direct Comparison Test (DCT)</u>

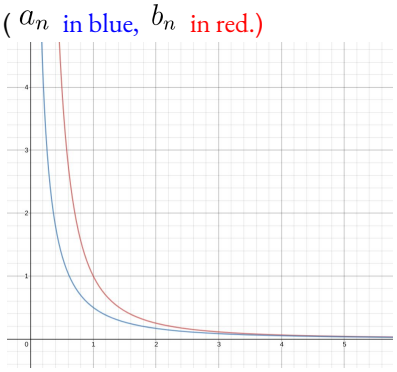
Say that we know $\sum a_n$ and $\sum b_n$ such that $\sum a_n \leq \sum b_n$ for all sufficiently large n (that is, as n approaches infinity). If we can prove that $\sum b_n$ converges, then it follows that $\sum a_n$ converges as well.

This should be fairly intuitive - we can almost think of $\sum b_n$ as an "upper bound" for $\sum a_n$ since $\sum b_n$ must eventually dominate $\sum a_n$. If $\sum b_n$ converges, since $\sum a_n$ is effectively "smaller" than $\sum ab$, then $\sum a_n$ must also converge.

Similarly, if $\sum a_n \leq \sum b_n$ for all sufficiently large n, and $\sum a_n$ diverges, then it follows that $\sum b_n$ must also diverge. In other words, if the "smaller" series diverges, then the "bigger" series must also diverge.

The diagram below captures the intuition used to justify the DCT.

(a_n in blue, b_n in red.)



(If red converges, then blue also converges. If blue diverges, then red also diverges.)

<u>We formally state this test as follows:</u>

Let $0 \leq a_n \leq b_n$ for all sufficiently large values n.

If $\sum b_n$ converges, then $\sum a_n$ also converges.

If $\sum a_n$ diverges, then $\sum b_n$ also diverges.

Let's look at a few examples:

1. $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ converges since $\sum_{n=1}^{\infty} \frac{1}{n^2+1} < \sum_{n=1}^{\infty} \frac{1}{n^2}$ for all n, and $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges by the Integral Test.

2. $\sum_{n=1}^{\infty} \frac{1}{n-\frac{1}{2}} \geq \sum_{n=1}^{\infty} \frac{1}{n}$ for all $n \in [1, \infty)$. Since we know that the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, we can conclude that by the Direct Comparison Test, $\sum_{n=1}^{\infty} \frac{1}{n-\frac{1}{2}}$ diverges as well.

<u>The Limit Comparison Test (LCT)</u>

Let $\lim_{n\to\infty} \frac{a_n}{b_n} = C$ with $0 < C < \infty$. Then either $\sum a_n$ and $\sum b_n$ both converge or diverge.

The intuition behind the Limit Comparison Test is fairly straightforward: If a_n and b_n have the same ratio, we can think of them as increasing at the same rate; just one is just a version of the other with a bigger magnitude. Nevertheless, they will share the same end behavior.

As usual, let's look through a few examples.

1. $\sum_{n=1}^{\infty} \frac{n^2+7n+6}{3n^3+1}$ diverges, since we know $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, and $\lim_{n\to\infty} \frac{n^2+7n+6}{3n^3+1} \cdot \frac{n}{1} = \frac{1}{3}$.
2. $\sum_{n=1}^{\infty} \frac{1}{n^2-n+6}$ behaves, on a large scale, similarly to $\sum_{n=1}^{\infty} \frac{1}{n^2}$. $\lim_{n\to\infty} \frac{n^2}{n^2-n+6} = 1$, and since 1 is on the necessitated interval, we know that by the Limit Comparison Test, $\sum_{n=1}^{\infty} \frac{1}{n^2-n+6}$ has the same end behavior as $\sum_{n=1}^{\infty} \frac{1}{n^2}$. Since $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges by the P-Series Test, $\sum_{n=1}^{\infty} \frac{1}{n^2-n+6}$ diverges also.

<u>Ratio Test</u>

Let $\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = C$, if it exists.

If $C < 1$, then $\sum a_n$ converges absolutely.

If $C > 1$, then $\sum a_n$ diverges.

If $C = 1$, the test is inconclusive.

Intuitively, if the ratio between the term after it and the term self is less than 1 when approaching infinity, the series converges. If it's greater than 1, then it diverges. This makes sense, since if its terms are decreasing every time (since they are being multiplied by a number less than 1), then its terms will eventually get to 0, meaning the sum should converge. If it's greater than 1, though, its terms will increase each time, meaning the sum must diverge because you're adding bigger and bigger things an infinite number of times.

More examples:

1. $\sum_{n=1}^{\infty} \frac{e^n}{n!}$ converges since $\lim_{n\to\infty} \left|\frac{e^{n+1}}{(n+1)!} \cdot \frac{n!}{e^n}\right| = \lim_{n\to\infty} \left|\frac{e^n(e)}{(n+1)n!} \cdot \frac{n!}{e^n}\right| = \lim_{n\to\infty} \left|\frac{e}{(n+1)}\right| = 0 < 1$.

2. $\sum_{n=1}^{\infty} \frac{n^n}{n!}$ is inconclusive since $\lim_{n\to\infty} \left|\frac{n^{n+1}}{(n+1)!} \cdot \frac{n!}{n^n}\right| = \lim_{n\to\infty} \left|\frac{n^n(n)}{(n+1)n!} \cdot \frac{n!}{n^n}\right| = \lim_{n\to\infty} \left|\frac{n}{(n+1)}\right| = |1|$.

This test works best for factorials and/or exponentials, as they usually can be cancelled out as in the previous 2 examples.

<u>The Root Test</u>

Let $\lim_{n\to\infty} \sqrt[n]{a_n} = C$, if it exists.

If $C < 1$, then $\sum a_n$ converges.

If $C > 1$, then $\sum a_n$ diverges.

If $C = 1$, the test is inconclusive.

A few examples:

1. $\sum_{n=1}^{\infty} \left(\frac{n}{3n+7}\right)^n$ converges since $\lim_{n\to\infty}\left(\frac{n}{3n+7}\right) = \frac{1}{3} < 1$.
2. $\sum_{n=1}^{\infty} \left(\frac{3n+4}{2n}\right)^n$ diverges since $\lim_{n\to\infty}\left(\frac{3n+4}{2n}\right) = \frac{3}{2} > 1$.

The intuition for this should be similar to the Ratio Test. If it's something less than 1 being raised to the nth power, the result will remain less than 1.

This is the probably the best test to use if the whole series can be written to a power of n.

<u>ALTERNATING SERIES:</u>

An *alternating series* is of the form $\sum_{n=1}^{\infty} (-1)^n a_n$. For example,

$\sum_{n=1}^{\infty} (-1)^n \frac{1}{n} = -1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} + \ldots + \frac{1}{n} + \ldots$ is an *alternating series*.

<u>Absolutely v. Conditionally Convergent</u>

If $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum_{n=1}^{\infty} a_n$ must also converge. In this case, the series $\sum_{n=1}^{\infty} a_n$ is said to *converge absolutely.*

If $\sum_{n=1}^{\infty} |a_n|$ diverges and $\sum_{n=1}^{\infty} a_n$ converges, then the series is said to *converge conditionally.*

For instance, $\sum_{n=1}^{\infty} |(-1)^n \frac{1}{n}| = \sum_{n=1}^{\infty} \frac{1}{n}$ which diverges, but $\sum_{n=1}^{\infty} (-1)^n \frac{1}{n}$ converges by the Alternating Series Test. As such, this series con*verges conditionally.*

<u>Alternating Series Test</u>

An alternating series of the form $\sum_{n=1}^{\infty} (-1)^n a_n$ converges if the following conditions are met:

$\lim_{n\to\infty} a_n = 0$

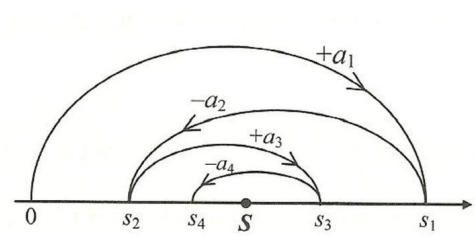
$a_{n+1} < a_n$ for all n.

1. $\sum_{n=1}^{\infty} (-1)^n \frac{1}{n}$ converges by the Alternating Series Test since

$\lim_{n\to\infty} |(-1)^n \frac{1}{n}| = \lim_{n\to\infty} |\frac{1}{n}| = 0$

$|\frac{1}{n+1}| < |\frac{1}{n}|$ for all n in the domain of $\{a_n\}$.

<u>Alternating Series Error Bound</u>



The error, or difference between the nth term of an alternating series and the limit, S, must be less than the next term in the series. That is,

$R_n < a_{n+1}$. The a_{n+1} term is called the *error bound*.

Sample Question: *How many terms must be summed to approximate to 3 decimal places the value of* $\sum_{n=1}^{\infty} (-1)^n \frac{1}{n^2}$?

Step 1:

$\lim_{n\to\infty} |(-1)^n \frac{1}{n^2}| = \lim_{n\to\infty} |\frac{1}{n^2}| = 0$

$|\frac{1}{(n+1)^2}| < |\frac{1}{n^2}|$ for all n,

$\sum_{n=1}^{\infty} (-1)^n \frac{1}{n^2}$ converges by the Alternating Series Test.

Step 2:

$R_n < \frac{1}{(n+1)^2} < 0.001.$

$(n+1)^2 > 1000$

$(n+1) > 10\sqrt{10}$

$(n > 10\sqrt{10} - 1 \approx 30.6$

31 terms must be summed to approximate the value of the series.